

Image tensors and channels

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can read a pixel using declared row column and channel axes.

I can convert channel layout without changing pixel meaning.

I can distinguish fixed value scaling from information losing transformations.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Read a pixel using declared row column and channel axes

Convert channel layout without changing pixel meaning

Distinguish fixed value scaling from information losing transformations

Retrieve what you already know

Why does a representation with two components not necessarily preserve two original measurements?

My starting explanation

This week's end goal

By the end, I can explain image tensors and channels using a traced example and a stated limitation.

2. Notice the evidence

A small fictional RGB image has height 2 pixels, width 2 pixels and three channels ordered red, green, blue. Store it as `image[row][column][channel]`, abbreviated HWC. Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are channels 0, 1 and 2. Each stored component is an integer from 0 to 255.

Position · RGB triplet · Meaning

Position: Row 0 column 0

RGB triplet: [255, 0, 0]

Meaning: Red

Position: Row 0 column 1

RGB triplet: [0, 255, 0]

Meaning: Green

Position: Row 1 column 0

RGB triplet: [0, 0, 255]

Meaning: Blue

Position: Row 1 column 1

RGB triplet: [255, 255, 255]

Meaning: White

The shape is 2 by 2 by 3. There are four pixels and twelve channel values. The blue component of the bottom-left pixel is `image[1][0][2] = 255`. A single channel is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three channels here; other formats can include additional channels. Declare the format instead of assuming every three-dimensional array means the same thing.

List the given facts and one fact that is not supplied.

3. See how it works

A channel-first layout uses `image[channel][row][column]`, abbreviated CHW. Our image then has shape 3 by 2 by 2. A correct conversion preserves the rule `chw[c][r][k] = hwc[r][k][c]`. The code assumes a nonempty rectangular image with exactly three values per pixel.

```
hwc = [
    [255, 0, 0], [0, 255, 0],
    [0, 0, 255], [255, 255, 255]],
]

height, width = len(hwc), len(hwc[0])

chw = [[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]

for channel in chw:
    print(channel)

assert chw[2][1][0] == hwc[1][0][2] == 255

restored = [[[chw[c][r][k] for c in range(3)]
              for k in range(width)] for r in range(height)]

assert restored == hwc
```

Reading nested values in order and merely regrouping them into a new shape does not perform this axis conversion. The first four flattened HWC values are `[255, 0, 0, 0]`. A naive regrouping makes these the first channel plane, but the correct red plane is `[[255, 0], [0, 255]]`. Shape agreement alone cannot detect this mistake.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Add a batch containing three identical images in NHWC layout.

Prediction: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour channel.

5. Words that unlock the idea

Word · Meaning

Word: Tensor

Meaning: A numerical array with a declared set of axes

Word: Channel

Meaning: One component plane of an image such as red green or blue

Word: Axis order

Meaning: The sequence of meanings assigned to array dimensions

Word: Value scaling

Meaning: A declared numerical conversion applied to stored values

Use two terms to explain a step in this week's example.

6. Follow a worked example

The same pixels in three channel planes

Channel · Top row · Bottom row

Channel: Red c 0

Top row: [255, 0]

Bottom row: [0, 255]

Channel: Green c 1

Top row: [0, 255]

Bottom row: [0, 255]

Channel: Blue c 2

Top row: [0, 0]

Bottom row: [255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image tensors and channels

Move colour values between tensor layouts without changing the pixels.

The facts we will use

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, channel.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.



2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].



3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Record HWC shape

What happens and why: Shape is [2, 1, 3]; top green is 4 and bottom green 5.

Step: 2. Reorder to CHW

What happens and why: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

Step: 3. Check same pixel

What happens and why: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A tensor is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour channel.

What this example does not establish

Shape consistency does not prove correct colour order or value range; document channel names and preprocessing.

Read the labelled picture: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Add a batch containing three identical images in NHWC layout. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

For this lesson, divide every channel value by 255 to obtain floating-point values from 0 to 1. Thus [255, 128, 0] becomes [1, 128/255, 0], with the middle value approximately 0.502. This fixed conversion does not fit a statistic from data. Apply the same conversion during training and later use, and record whether an input is already scaled.

Dividing twice makes a raw value of 255 become $1/255$, approximately 0.00392, instead of 1. Integer rounding after scaling can also erase distinctions. In exact arithmetic the declared division is reversible by multiplication; finite precision or deliberate rounding can limit exact recovery.

Averaging RGB channels is a different operation. The classroom mean gives red [255, 0, 0] and green [0, 255, 0] the same value 85, losing their colour distinction. It is a simple illustration, not a perceptual brightness formula. If preprocessing instead estimates means or spreads from a dataset, fit those statistics using training data only.

Locate the earliest unsupported step

Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

State one repair and the checks you would rerun.

8. Practise together

For the lesson image, give `image[1][0]`, `image[1][0][2]`, the HWC shape, pixel count and number of channel values.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

Explain your prediction to a partner before comparing results.

Practise with less help: Audit an axis conversion

Use workbook W2 for the connected example “Image tensors and channels”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image tensors and channels. Proposed explanation: “Three channels mean three images”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

```
hwc = [
    [[255, 0, 0], [0, 255, 0]],
    [[0, 0, 255], [255, 255, 255]],
]
height, width = len(hwc), len(hwc[0])
chw = [[hwc[r][k][c] for k in range(width)]
        for r in range(height)] for c in range(3)]
for channel in chw:
```

```

print(channel)

assert chw[2][1][0] == hwc[1][0][2] == 255

restored = [[chw[c][r][k] for c in range(3)]
             for k in range(width)] for r in range(height)]

assert restored == hwc

```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For the lesson image, give `image[1][0]`, `image[1][0][2]`, the HWC shape, pixel count and number of channel values.

Scale `[255, 128, 0]` by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled image tensors and channels evidence sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and channel axes. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

Stacking five same-size HWC images adds a leading example axis N. This batch has shape 5 by 2 by 2 by 3 and contains 60 channel values. With channel-first images, its shape is 5 by 3 by 2 by 2. The example axis is not another colour channel. Every axis must keep its meaning through the model.

A loader returns the expected shape, but red and blue channels were exchanged and already scaled values were divided by 255 again. Does the shape check establish correctness? Name an exact pixel check and a value check that expose these faults.

Use a tiny known image, named axis meanings and expected channel values to check a pipeline. Keep original images when testing transformations. Flattening can be reversed if shape and traversal order are retained; averaging colours generally cannot. Next week reads small neighbourhoods with a convolution operation.

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

The same pixels in three channel planes

Channel · Top row · Bottom row

Channel: Red c 0

Top row: [255, 0]

Bottom row: [0, 255]

Channel: Green c 1

Top row: [0, 255]

Bottom row: [0, 255]

Channel: Blue c 2

Top row: [0, 0]

Bottom row: [255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Convert a new image independently

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all

three CHW planes. Give the green value at row 0 column 1 in both layouts. A batch contains four copies: give its NHWC shape and total channel-value count.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts. A batch contains four copies: give its NHWC shape and total channel-value count.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Move colour values between tensor layouts without changing the pixels.

By the end, I can · How I will show it

By the end, I can: Read a pixel using declared row column and channel axes

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Convert channel layout without changing pixel meaning

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Distinguish fixed value scaling from information losing transformations

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Read a pixel using declared row column and channel axes

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Move colour values between tensor layouts without changing the pixels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

1. Record HWC shape
2. Reorder to CHW
3. Check same pixel

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Check same pixel Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. A tensor is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour channel.

Practice 2: explain the deciding evidence

Changed-case practice: Add a batch containing three identical images in NHWC layout. Which conclusion is supported by the supplied facts?

1. Batch shape is [3, 2, 1, 3].
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour channel.

Repair a missing building block

Review Learned representations (ai-5/19). Goal: Trace an input through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed training budget. Short practice: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.